

GENAI FOR TEST MANAGEMENT • TRACK 2

Module 3

PR Scope, Coverage-Gap Checks & Responsible AI Governance

The fix in this module's PR (ticket AEB-99) closes the same defect first caught in Module 1's Sprint 14 retro (QAM-207) — still open in Module 2's Sprint 15 backlog as QAM-315. One thread, three modules.

Slides 1–9 • Hands-on lab included • ~60 min

Checking PR Scope Against Requirements with Copilot

PR #482 vs. ticket AEB-99 — does the fix match the ticket's scope?

sensorFusion.c — Validate radar return vs. camera-confirmed range-rate; fall back to camera-only on failure

IN SCOPE

brakeActuator.c — Brake ramp time shortened 400ms → 180ms, cited as "NCAP scoring guidance"

OUT OF SCOPE

aebStatusFrame.c — Added ttc_estimate_ms to the AEB status CAN frame for the infotainment team

OUT OF SCOPE

legacyRadarCanMap.c — Deleted the legacy radar hardware's CAN ID mapping entirely

OUT OF SCOPE



Highest operational risk

The CAN schema change (aebStatusFrame.c) is the riskiest item despite sounding benign — it silently changes a message contract another ECU (infotainment) depends on. All three out-of-scope changes are individually defensible-sounding; ticket AEB-99 explicitly excludes all three in its own "Out of scope" section.

Requirement vs. Test-Result Coverage Check

Same skill as Slide 2, applied to requirements vs. this build's test-execution results

Requirement ID	Title	Priority	Status
AEB-99	Radar plausibility validation	Critical	Pass
AEB-101	TTC re-evaluation after lock loss	High	Pass
AEB-105	Pre-collision warning latency	Medium	Failed
AEB-108	Graceful degradation under sensor overload	Critical	Failed
AEB-114	TTC recalculation on changing deceleration	Critical	No Test — Gap



"No test" and "test failed" are different problems

AEB-114 has zero coverage on a requirement that maps to an already-open Critical defect (QAM-306, Module 2's sandbox) — arguably the most urgent finding here, and it wouldn't show up at all in a report that only looks at pass/fail results. In ASPICE-assessed environments this is the same distinction requirements-traceability reviews are built to catch.

Why "Advisory-Only" Is the Right Default

Fairness

Reliability & Safety

Privacy & Security

Inclusiveness

Transparency

Accountability

GitHub: Copilot "may not identify all of the problems... especially where changes are large or complex."

i

In ISO 26262 environments

Keeping AI output strictly advisory — never feeding it directly into a safety case or test-evidence package — is also what keeps it outside the standard's tool-qualification requirement (ISO 26262-8, clause 11: a tool needs qualification, at Tool Confidence Level 2 or 3, only if it could inject or fail to catch an error affecting a safety-relevant work product).

Responsible AI Use From a QA Leadership Seat

1

External-facing or compliance artifact?

→ Mandatory review

2

Sensitive or proprietary data — OEM-proprietary requirements, ASIL-classified specs, customer PII?

→ Check your NDA/data-handling policy before pasting into any prompt

3

One-off draft vs. recurring automation?

→ Recurring needs a periodic audit of the automation itself

AI Governance and Ethics for QA

1

Fairness

Don't replace fair performance conversations with AI-flagged "risks."

2

Transparency

Disclose when a report was AI-assisted.

3

Accountability

A human always signs off.

4

Data handling

Synthetic/sanitized by default; real production data only under existing org policy.

Validating AI Output: Reviewing Against a Source of Truth

- Compare AI output line-by-line against something already known to be correct
- Scope check: re-open the ticket and check flagged items yourself
- Coverage-gap list: compare against the answer key — note false positives and false negatives

AI SAID

"3 coverage gaps: AEB-105, AEB-108, AEB-114" — treated as equivalent findings.

ANSWER KEY SAYS

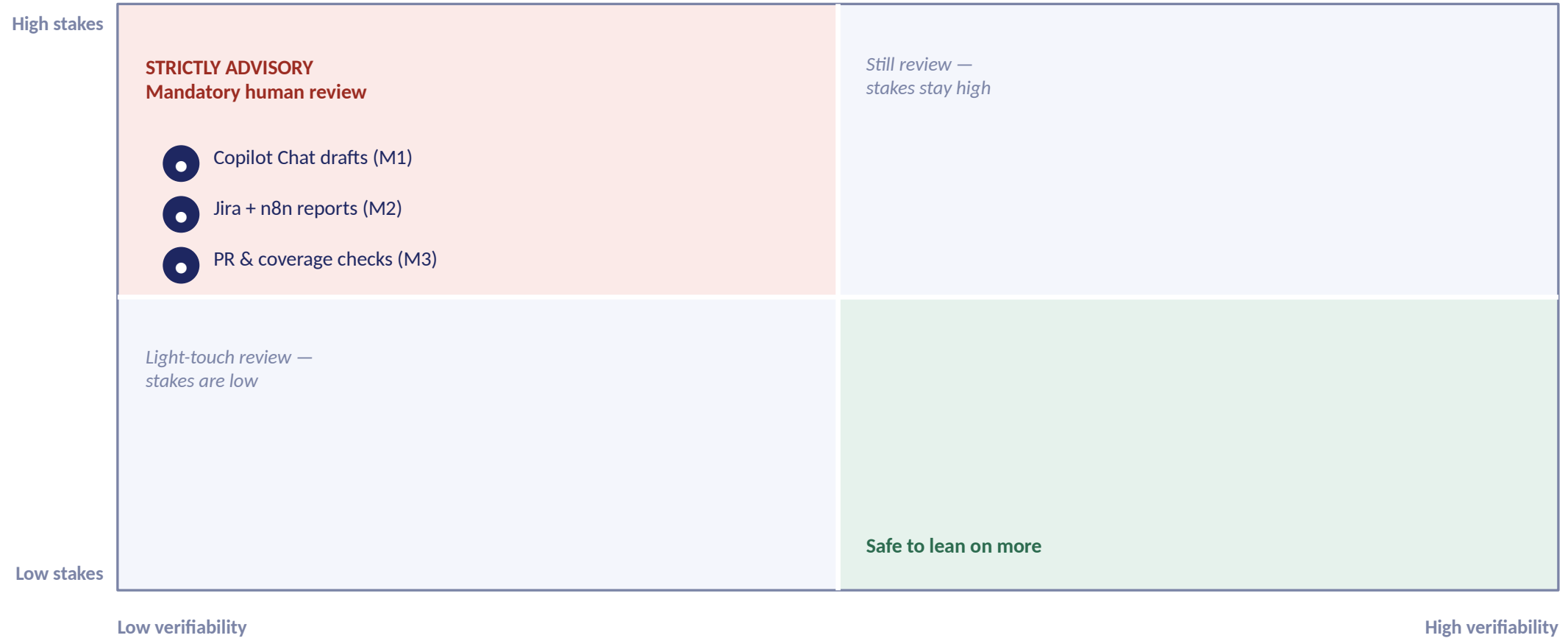
AEB-105 & AEB-108: tests exist but FAILED. AEB-114: NO test exists — and maps to open Critical defect QAM-306. Not the same problem.



This is a confirmation review

The same review discipline higher-ASIL work products already require — just applied to an AI-assisted first pass instead of a human-only one.

Recognizing When to Keep a Tool Advisory-Only



Module 3 Lab



Task A: PR scope match assessment (Slide 2)



Task B: AEB requirement vs. test-result coverage-gap list (Slide 3)



Task C: Review your own result against the answer key (Slide 7)



Lab data ready

ticket-AEB-99.md · pr-diff.md · aeb-requirements.csv · test-execution-report.xml/.html · execution-log.txt · answer-keys/pr-scope-answer-key.md · answer-keys/coverage-gap-answer-key.md — all in lab/module-3-scope-coverage/. Runs fully offline, ~60 min including the governance discussion.